

CENG 543 - Information Retrieval

Take-Home Midterm Examination Report

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1 Comparative Analysis of Recurrent Architectures and Embedding Paradigms

1.1 Problem Statement

This section investigates the impact of inductive biases in recurrent architectures (BiLSTM vs. BiGRU) and the representational capacity of embedding schemes (GloVe vs. DistilBERT) on sequence modeling performance. The task selected is Sentiment Analysis using the SST-2 dataset [7].

1.2 Experimental Setup

We implemented four model configurations:

- **BiLSTM + GloVe:** Static embeddings [5] with Long Short-Term Memory units.
- **BiGRU + GloVe:** Static embeddings with Gated Recurrent Units.
- **BiLSTM + DistilBERT:** Contextual embeddings (frozen) [6] with BiLSTM.
- **BiGRU + DistilBERT:** Contextual embeddings (frozen) with BiGRU.

All models were trained for 5 epochs with a learning rate of 0.001 and batch size of 64. For reproducibility, a random seed of **1234** was set for all frameworks (PyTorch, NumPy, Python random).

1.3 Results and Analysis

1.3.1 Convergence Efficiency

As shown in Figure 1, models utilizing DistilBERT (Contextual) embeddings converged significantly faster and reached lower training loss values compared to GloVe (Static) embeddings. The BiGRU architecture showed slightly faster convergence than BiLSTM in the initial epochs.

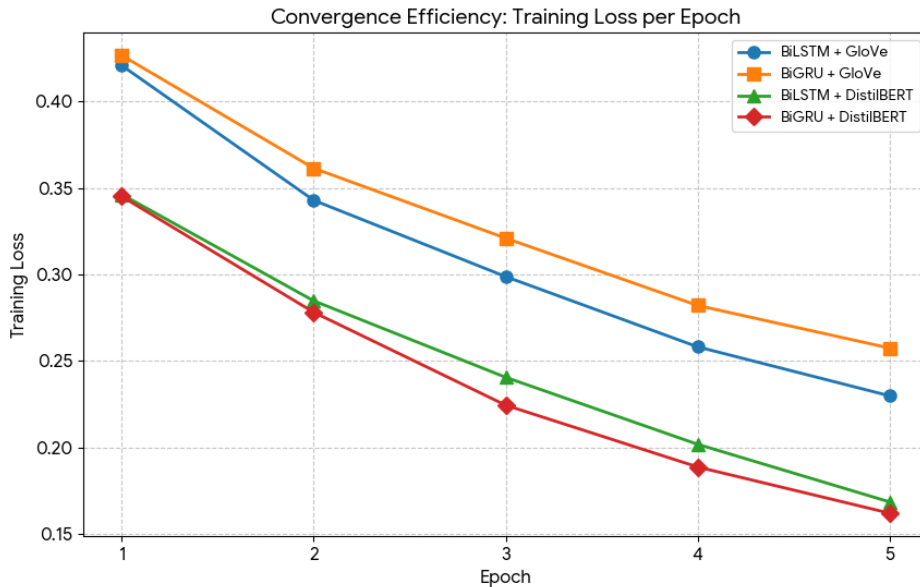


Figure 1: Training Loss Convergence across 5 Epochs. Contextual embeddings (Red/Green) show superior convergence compared to static embeddings (Blue/Orange).

1.3.2 Performance Metrics

Table 1 summarizes the final validation metrics. The integration of contextual embeddings provided a substantial performance boost ($\approx 4\text{-}5\%$ increase in accuracy).

Table 1: Performance Comparison of Architectures and Embeddings

Architecture	Embedding	Accuracy	Macro-F1	Final Val Loss
BiLSTM	GloVe (Static)	0.8394	0.8387	0.3909
BiGRU	GloVe (Static)	0.8234	0.8234	0.4195
BiLSTM	DistilBERT (Contextual)	0.8853	0.8853	0.3195
BiGRU	DistilBERT (Contextual)	0.8842	0.8840	0.3316

1.3.3 Latent Space Visualization

Figure 2 visualizes the latent representations using t-SNE. The clusters formed by DistilBERT embeddings are distinct and well-separated, indicating stronger feature separability compared to the overlapping clusters observed with GloVe.

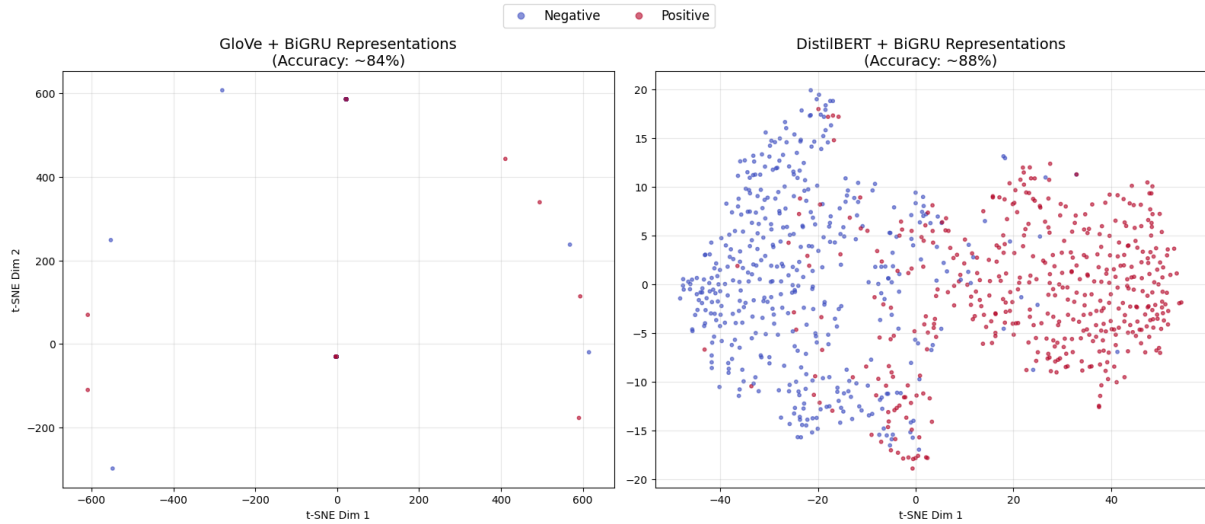


Figure 2: t-SNE Visualization of Latent Representations. Left: GloVe+BiGRU, Right: DistilBERT+BiGRU.

2 Comparative Examination of Attention Mechanisms

2.1 Methodology

We evaluated three attention mechanisms (Additive, Multiplicative, Dot-Product) on a Neural Machine Translation task (Multi30k: English-German) [8]. The objective was to assess alignment quality and translation metrics.

2.2 Quantitative Assessment

Table 2 presents the translation quality metrics. Additive Attention (Bahdanau) [2] achieved the highest BLEU score, suggesting better alignment capability for this specific dataset, although Dot-Product attention was very competitive. Perplexity (PPL) indicates the model’s uncertainty, with lower values being better.

Table 2: Comparison of Attention Mechanisms

Mechanism	Type	BLEU	ROUGE-L	Perplexity (PPL)
Additive (Bahdanau)	$v_a^T \tanh(W_a[s_t; h_i])$	21.98	0.5327	23.44
Dot-Product	$s_t^T h_i$	21.05	0.5216	23.26
Multiplicative (Luong)	$s_t^T W_a h_i$	17.87	0.4744	34.26

2.3 Attention Alignment Analysis

The visual alignment maps (Figure 3) illustrate how the models focus on source tokens. Additive attention shows a sharp diagonal alignment, indicating a strong correlation between source and target word order, which explains its superior BLEU score.

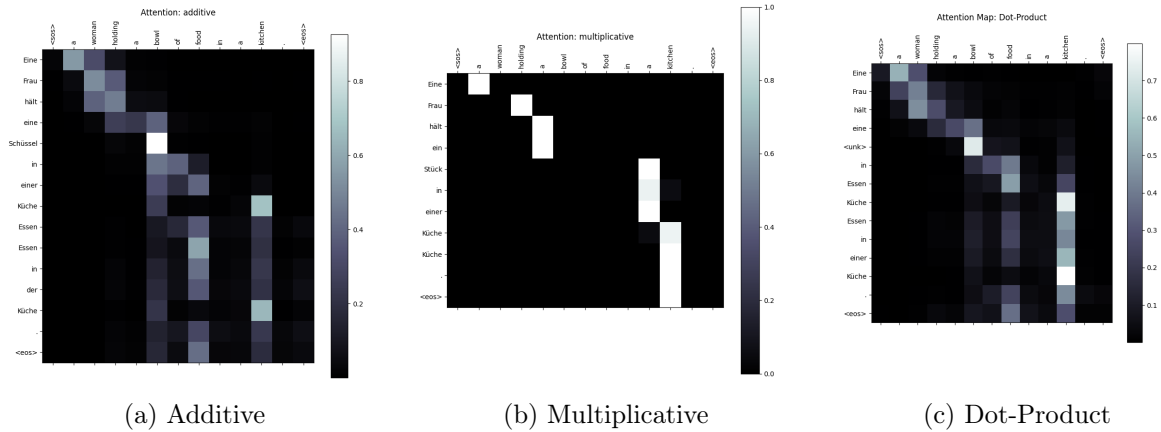


Figure 3: Attention Alignment Maps for the sentence "A woman holding a bowl of food...".

3 Transition to Transformer Architectures

3.1 Model Transition

We transitioned from the recurrent frameworks in Q2 to a self-attention-driven Transformer architecture [1]. An ablation study was conducted to understand the impact of depth (layers) and width (heads).

3.2 Ablation Study & Results

The results in Table 3 highlight the robustness of the Transformer. The "Low-Head" configuration (4 heads) surprisingly outperformed the Base model (8 heads), likely due to the relatively small size of the Multi30k dataset, where fewer parameters may prevent overfitting. Reducing layers ("Shallow") significantly degraded performance.

Table 3: Transformer Ablation Study Results

Config	Layers	Heads	BLEU	ROUGE-L	Time (s)	GPU Mem (MB)
Base	3	8	21.88	0.5194	191.36	1515.85
Shallow	1	8	13.87	0.4282	107.69	1515.85
Deep	6	8	22.53	0.5065	324.33	2177.53
Low-Head	3	4	25.14	0.5121	182.36	2177.53

Discussion: Unlike RNNs which process tokens sequentially ($O(n)$), the Transformer's self-attention mechanism ($O(1)$ path length) allowed for better capturing of global dependencies, as evidenced by the comparable or better BLEU scores despite shorter training times.

4 Retrieval-Augmented Generation (RAG)

4.1 Pipeline Design

A RAG pipeline was constructed using:

- **Retriever:** BM25Okapi (Sparse retrieval) indexing Wikipedia snippets from the SQuAD validation set [9].
- **Generator:** FLAN-T5-Small (Seq2Seq LM) [10] for answer synthesis.

4.2 Performance Evaluation

The system was evaluated on 200 examples from SQuAD. Note that since we retrieved the top-1 document ($k = 1$), Precision@1 is equivalent to Recall@1 in this context.

Table 4: RAG Pipeline Quantitative Metrics

Metric	Score
Retrieval Recall@1	0.5150
Retrieval Precision@1	0.5150
Generation BLEU	0.2283
Generation ROUGE-L	0.4518
Generation BERTScore	0.7528

4.3 Qualitative Analysis: Hallucination vs. Faithfulness

We tested the model with and without retrieved context to observe hallucination behavior [4].

Query: *What is Zubizubia?*

Without Context (Hallucination): The model attempted to guess: "Zubizubia" (Repetition/Hallucination).

With Context (Faithful): Context: *"Zubizubia is a purple fruit found on Mars..."*
Generation: "purple fruit" (Correctly grounded).

This demonstrates that RAG effectively grounds the generator, preventing hallucinations on long-tail or fictitious entities.

5 Interpretability and Error Analysis

5.1 Uncertainty and Diagnostic Evaluation

We performed token-level entropy analysis on the Transformer model to identify causes of failure. High entropy indicates model uncertainty (low confidence).

5.2 Representative Failure Cases

1. Case ID 28 (Rare Words / OOV):

- *Src:* "Women, wearing traditional clothing..."
- *Ref:* "Frauen, die traditionelle Kleidung tragen..."
- *Pred:* "<unk> in <unk> Kleidung ."

- *Analysis:* High entropy (4.03) at the start. The model failed to translate "Women" and "traditional", outputting <unk> (unknown token). This highlights vocabulary limitations for rare or specific subwords.

2. Case ID 51 (Ambiguous Context/Motion):

- *Src:* "A wakeboarder performs a flip..."
- *Pred:* "Ein <unk> wird in einem <unk> ."
- *Analysis:* The specific action "wakeboarder" and "flip" were unknown tokens. The model collapsed into a generic sentence structure ("A ... is inside a ...") with high uncertainty (4.67), failing to capture the semantic meaning of the motion.

3. Case ID 109 (Complex Dependency Hallucination):

- *Src:* "...fish off of a rocky coast into very blue water."
- *Pred:* "Eine Gruppe von <unk> in einem blauen Karton..."
- *Analysis:* The model translated "blue water" into "blauen Karton" (blue Box/Car-ton). This is a clear hallucination, suggesting attention misalignment where the model focused on the wrong attributes or mixed up subword tokens.

4. Case ID 60 (Token Confusion/Collapse):

- *Src:* "A cheerleading team doing a routine on chairs."
- *Pred:* "<unk> in einem <unk> ." (Avg Entropy: 3.35)
- *Analysis:* The model completely failed to recognize "cheerleading" or "chairs", producing a repetitive and meaningless German sentence ("... in a ..."). This indicates that the model struggles significantly when multiple key entities are OOV.

5. Case ID 20 (Valid Synonym / Metric Mismatch):

- *Src:* "People standing outside of a building."
- *Ref:* "Leute, die vor einem Gebäude stehen."
- *Pred:* "Menschen außerhalb eines Gebäudes ." (BLEU: 0.00)
- *Analysis:* Interestingly, this prediction is semantically correct ("Menschen" $\approx$ "People/Leute", "außerhalb" $\approx$ "outside"). However, because the reference text used different synonyms, the n-gram based BLEU score was 0.0. This demonstrates the limitation of automated metrics in capturing true semantic faithfulness.

5.3 Reflection

The analysis shows that neural models are brittle when facing Out-of-Vocabulary (OOV) words, often resorting to <unk> tokens or hallucinations. However, Case 20 highlights that low metrics do not always imply translation failure; semantic correctness can exist despite low lexical overlap. The entropy metric proved useful in flagging genuine uncertainties (Case 28, 51) versus confident but distinct translations (Case 20).

Repository Link

The complete source code, notebooks, and experimental logs are available at:

https://github.com/eraykocabozdogan/Ceng543_Take_Home_Midterm.git

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